SPACE: The Unfinished Universe

Wormhole 01 — Asymmetry, A Cloud School, by Naxxatra · Oct 16, 2026

1. Executive summary

This document designs Wormhole 01 for Space: a seven-minute episode, a Reply that doubles as an application, and a map of the longer journey behind it. Its central move is to make the learner's first task the same one the fictional civilisation fails at: deciding what a signal actually shows before deciding what it means.

The design in five lines

1. **Setting:** a Planet of Polymaths, a civilisation fluent in every field, that receives an unexplained signal from a distant star.
2. **Turn:** a young observer asks whether the signal is a message at all. Nobody has checked. The civilisation's intelligence did not protect it from an untested assumption.
3. **Tool:** first-principles thinking (the Thinking Tool assigned to Space in the brief), taught as an eight-step sequence, not as "keep asking why".
4. **Reality:** the fiction hands over to real episodes where scientists faced the same fork: the 1967 pulsar discovery, the Parkes "perytons", and the 2019 BLC1 candidate. A working scientist appears on screen here, as the brief requires.
5. **Mission:** the viewer picks a side in two minutes (message or not), then sends a Reply proposing how to find out. The Reply is the application.

Where this document aligns with, or adjusts, the Cloud School brief

Topic	What the brief says	What this document does
Episode length	6:30–7:30 minutes	Targets 7:00, with a scientist on camera in Act 5
Application	A Reply, scored on curiosity, reasoning, honesty and agency, never on polish, accent or English fluency	Uses those four criteria publicly. The six areas in the master prompt become reviewer prompts beneath them (section 10)

Topic	What the brief says	What this document does
Entry path	Watch, then Spark ("pick a side in two minutes"), then Reply	Designs the Spark explicitly: a two-minute lean toward "message" or "not a message"
Language	Terms to use: Wormhole, Studio, Reply, Residency, Fellow, Explorer, Thinking Tool	Used throughout. "Mission" appears only as the name of the task inside a Reply Card
Open-ended unknowns	"Nobody knows yet" is said proudly	The signal stays unresolved. So do both ultimate questions

Three things to decide first (detail in section 15): which working scientist fronts Act 5; whether to publish the title "The Signal That Shouldn't Exist" as is (recommended, with a payoff line in the film); and whether the Hindi version is dubbed or separately written (recommended: separately written, because the narration leans on idiom).

2. The identity of the Space Wormhole

The Space Wormhole is a seven-minute story about a signal, built so that a 10–17-year-old finishes it wanting to check something rather than wanting to know the answer.

Context: what Asymmetry is for

From the brief, in its own terms: Asymmetry, A Cloud School is an ecosystem where 10–17-year-olds work on the biggest open questions facing humanity, through weekly live Studios, in-person Innovation Residencies and public presentation at Frontiers. It grows three kinds of people (Polymath, Researcher, Innovator) around one shared core, high agency. Eight Wormholes, each with one Thinking Tool, are the front door. Every Wormhole does three jobs, and this one does them as follows.

Job (from the brief)	How the Space Wormhole does it
Doorway into the curriculum	One signal opens astronomy, statistics, signal processing, music and pattern, astrobiology, engineering and ethics. Each later Studio can enter through one of those doors

Job (from the brief)	How the Space Wormhole does it
Window for the world	A free, public film that works with no prior knowledge, no jargon, and no sign-up. A 12-year-old, a parent and a teacher can all argue about it afterwards
Invitation to join	The film ends on an unfinished investigation. The Reply Card is the application: it shows how a child reasons, not what they have memorised

What is the school's commitment, and what is this document's proposal

- **Stated by the brief:** the Wormhole format (6:30–7:30 minutes, puzzle then real scientist then mission), the Thinking Tool for Space (first principles), the unifying lens (how do we know what is true?), the Reply scored on curiosity, reasoning, honesty and agency, a scientist answering three Replies on video after two weeks, English and Hindi at launch, low-bandwidth access.
- **Proposed here:** the Polymath Planet, the signal scenario, the eight-step sequence, the storyboard, the Reply Card, the seven-planet network, the culminating project. Nothing about the school's operations, pricing or staffing is assumed beyond the brief.

Title

Option	Verdict
SPACE: The Unfinished Universe	**Recommended.** Says the universe is incomplete without saying it is empty. "Unfinished" invites a 10-year-old to help. It echoes the brief's own register: nobody knows yet
SPACE: Nobody Knows Yet	Closest to brand voice, but works better as a tagline than a title
SPACE: What We Haven't Found	Reads as treasure hunt, narrower than the two ultimate questions
SPACE: The Question Between the Stars	Pretty, but vague about what the child does

For the episode, keep **The Signal That Shouldn't Exist**. It reads as a hook, but the film turns it: the signal exists, it was measured. What should not exist is certainty nobody has earned. The closing narration pays this off directly (section 8).

Identity at a glance

Element	Statement
One-sentence premise	A civilisation of polymaths receives a signal it cannot read, and a child asks whether there is anything to read at all
Central question	How do we know what is true when the evidence is strange and everyone is clever?
Mission statement	Join the investigation. Work out what the evidence can and cannot tell us, and propose how to find out more
Educational objective	By the end of the Wormhole path, a learner can separate observation from interpretation, name their assumptions, offer competing explanations, and propose a test that would tell them apart
Emotional experience	Wonder at the first image, unease when the experts disagree, a small jolt of recognition at the child's question, then the pleasure of being handed the problem
Intellectual experience	Watching a smart answer fall apart, then watching a better question rebuild it. Holding two explanations at once without rushing to pick
Longer journey	Space Studios on life, signals, settlements and ethics, a Dark Sky Expedition (May, per the brief), and a Frontiers research track
What it is not	A tour of planets and stars, a lecture on SETI, or an advert. It is a reasoning problem wearing a story

What makes it different from a space-science course

A course asks what is out there. This Wormhole asks how anyone would know. The facts (what a pulsar is, what a radio telescope measures) arrive only when the argument needs them, and the child is judged on how they handle uncertainty, not how many facts they hold. Space is the setting, first-principles thinking is the tool, and **how do we know what is true?** is the lens.

3. Mission, objectives and guiding question

Guiding question: How do we know what is true? Space is where that question is hardest, because we cannot touch, visit or repeat most of what we study.

Mission of the Wormhole: to make a child unable to leave the question *what lives in the deep universe, and what does leaving Earth mean for humanity?* alone, and to give them the first tool for working on it.

Learning objectives

These are written as things a learner can do, and apply across the full Wormhole path, not just the film. Each lines up with a Thinking Tool move (section 7) and with a Reply criterion (section 10).

#	A learner can…	Appears first in
1	Say what was observed, separately from what is believed about it	Episode, Act 4
2	Name at least one assumption hiding inside a confident explanation	Episode, Act 4
3	Offer two or more explanations that fit the same evidence	Reply Card
4	Describe a measurement that would favour one explanation over another	Reply Card
5	Say what would change their mind	Reply Card, Studio
6	Tell established science, plausible speculation and invention apart	Episode, Act 5

#	A learner can…	Appears first in
7	Weigh a technical possibility against whether it is desirable	Beyond-Earth project

What we want a learner to become

Not an astrophysicist. Someone who, handed a strange claim, asks what was measured, what is assumed and what would settle it, and who is comfortable saying "we do not know yet" without giving up. Different learners will get there differently: some by drawing, some by arguing, some by building. The path does not need to converge, and neither do their conclusions.

4. The two ultimate questions

Both questions come straight from the brief. Both stay open investigations. The Wormhole never implies that life exists elsewhere, that humanity must leave Earth, or that more technology means a better civilisation.

Question A: What lives in the deep universe?

Strand	What is solid	What nobody knows yet
Origin of life	Life on Earth appeared early in the planet's history and is chemically and genetically related	How non-living chemistry became life, and whether that is easy or rare
Habitability vs life	Habitable means conditions that could allow liquid water, not that anything lives there	Whether habitable worlds are usually inhabited
Extreme environments	Life on Earth survives boiling vents, Antarctic ice, deep rock, high radiation	Where the true limits lie, and whether other biochemistries could exist
Intelligence	Intelligence on Earth evolved once to the point of technology, in one species, after billions of years	Whether that is typical, rare or a fluke

Strand	What is solid	What nobody knows yet
Biosignatures and technosignatures	Scientists look for gases, surface patterns, radio signals, heat, unusual light	Which signs are reliable, since many can be mimicked by non-living processes
Anomaly vs evidence	An unusual detection starts an investigation, not a conclusion	Why, after decades, the question is still open

The last row is the Wormhole's centre of gravity. The hardest part of the search is not detecting something odd. It is deciding what an odd detection is allowed to mean.

Question B: What does leaving Earth mean for humanity?

Strand	What is solid	What nobody knows yet
Why leave	Reasons offered include scientific access, resilience, resources and curiosity	Whether any of these justify settlement, and who decides
Physical limits	Radiation, microgravity, closed-loop life support, distance and supply are real constraints	How much can be solved, at what cost, over what timescale
Visit vs sustain	A crew can live on a station with regular resupply	Whether a self-sustaining settlement of any size can be built and survive generations
Ethics and law	International space law and planetary protection principles exist	How they should apply to settlement, mining and possible indigenous life
Responsibility	Humans have affected every ecosystem they entered	What we owe other worlds, and what Earth can learn from the attempt

How the two questions connect

They meet in one point: if we go, what do we owe what we find, and what do we owe each other? A settlement could contaminate a world where life might exist, or find a world where it does. Question A shapes the rules for Question B, and Question B decides who is

looking for Question A's answer. The culminating project (section 13) forces them into one design.

5. Backward-designed learning journey

Working backwards from the two questions, a learner needs nine capabilities in order. Each depends on the one before; skip one and the next one collapses into opinion.

Stage	Why it is necessary	A learner can…	Where it first lives
1. Imagination	Without a world worth caring about, there is nothing to question	Picture a civilisation and a problem inside it	Episode, Acts 1–2
2. Questioning	Questions come before methods	Ask something the story did not answer	Episode, Act 3; Spark
3. Deconstruction	Strange claims have hidden parts	Split a claim into what was seen and what was assumed	Episode, Act 4
4. Evidence	Not all evidence is equal	Judge how strongly a measurement supports an idea	Episode, Act 5; Reply Card
5. Scientific models	Evidence needs explanations that can be tested	Offer two or more models that fit the same data	Reply Card; first Studio
6. Investigation	Explanations must be tested, not admired	Design a test that separates models	Reply Card; Studio
7. Design	Reaching places needs things that work	Turn constraints into a working proposal or prototype	Technology-linked Studios; Frontier Camp
8. Ethical reflection	What can be done and what should be done differ	Weigh benefits, harms and who decides	Planet of Justice; beyond-Earth project
9. Synthesis	Real questions cross disciplines	Defend a position that draws on all of the above	Culminating project; Frontiers

Why imagination comes first

A fictional planet lets a child feel an argument before learning its vocabulary. It also lowers the cost of being wrong: nobody is embarrassed about misjudging a planet that does not exist.

Progression by age division

The brief divides learners into Explorers (10–13) and Fellows (14–17). The same stages repeat at greater depth, not different content.

Stage	Explorer (10–13)	Fellow (14–17)
Deconstruction	Sort claims into "seen" and "believed"	Quantify uncertainty, separate systematic from random error
Evidence	Compare strong and weak clues	Read a real data plot, discuss false positives
Models	Two stories that fit the same clue	Two testable hypotheses with different predictions
Investigation	A test they could do with a phone, a ruler or a library	A proposal with a variable, a control and a failure condition
Ethics	Argue both sides of a rule	Draft and defend a policy with trade-offs

What the path does not assume

It does not assume one route, one conclusion or one discipline. A learner can reach stage 6 through a story, a model, a drawing or an argument. They can finish the Wormhole believing settlement is wise, unwise or conditional on things we have not solved.

6. The Polymath Planet and the Polymath Paradox

The Polymath Planet is a thought experiment, not a claim about any real world. Every feature below earns its place by producing a question, exposing an assumption or posing a problem a learner can work on.

Working name: Anvaya (a Sanskrit word for "connection"). Optional. If a more neutral name tests better with children, the world can stay "the Polymath Planet".

The world

Setting. A rocky, ocean-and-continent planet orbiting a long-lived orange star, with two moons. Its night sky is crowded, because the home star sits in an old, dense region of its galaxy. The far side of the larger moon, shielded from the planet's own radio noise, hosts the civilisation's main radio observatory, called **the Quiet Side**. Real radio astronomers also seek radio-quiet sites; this detail makes "interference" a natural question.

The people. Anvayans do not specialise. A child studies mathematics, music, biology, engineering, art and philosophy together, and a profession is a temporary project, not an identity. A musician can design a spacecraft; an artist can build a model of a storm; a child can challenge the planet's most celebrated scientist, and is expected to.

Energy. The civilisation gathers most of the energy that reaches its surface, and has begun placing collectors in orbit. On the Kardashev scale, which ranks civilisations by how much energy they command (planetary, stellar, galactic), they sit near the top of the first step. The scale measures power, not wisdom. Anvayans are formidable and not necessarily right.

How the design details earn their place

World detail	What it does for the learner
Observatory on a radio-quiet moon	Raises the question: how do you know your own instrument is not the source?
Cities built as laboratories open to children	Makes the polymath habit visible without a lecture
Knowledge shared through a public "Commons" where any claim can be challenged	Shows what a healthy culture of doubt looks like, then reveals where it fails
Art treated as a way of measuring (a mural that plots a star's brightness; a song that encodes a dataset)	Shows how representation changes what you notice, and what it can mislead you into seeing
Decisions made by Councils of mixed disciplines	Sets up the paradox: many experts, one shared blind spot
A grid that powers the planet yet cannot agree on how to share its output	Seeds the Planet of Invention and Planet of Justice, and the settlement questions in section 13

What the civilisation values, misunderstands and cannot solve

- **Values:** synthesis, beauty, evidence shown in public, being challenged by anyone.
- **Misunderstands:** that being able to explain a signal in many fields is the same as having established what it is. Each discipline finds its own pattern.
- **Cannot solve:** distance (the star is far; every reply takes years), unequal sharing of its own energy, and the habit of choosing the most interesting explanation first.

The signal

On the Quiet Side, a receiver records a burst from a star called **Vetra**. The pulses arrive in a sequence whose spacing looks mathematical for the first stretch, then does not. It arrives three times, at uneven intervals, and has been seen by one instrument only. The star is known to vary in brightness. The civilisation has no other cases to compare.

This detail is chosen so that no answer is forced. Interference, a natural source, an instrument fault and a deliberate signal each fit part of the data.

The Polymath Paradox

The Council of Anvaya spends many cycles interpreting the signal. The mathematician sees a numerical sequence. The musician hears rhythm and melody. The physicist models a spinning star. The linguist drafts translations. All four are brilliant, and all four have already assumed the signal is meaningful, which is a conclusion, not an observation.

The paradox: the more disciplines a mind commands, the more ways it can find a pattern, and the harder it becomes to notice that a pattern is not yet evidence. Breadth gives many tools for interpretation, and no instruction on when interpretation should begin.

The young observer

Ira, aged twelve, is an apprentice on the night shift who maintains the Quiet Side's calibration logs. She has no hypothesis about what the signal says. She asks what no decoder was built to ask:

> *What if we assumed it was a message before we checked that it was one?*

Ira is not right, and she is not wrong. She proposes a starting point: list what the instrument actually recorded, then ask which explanations that list allows. This is plausible and unglamorous. The adults do not instantly concede. Several object, reasonably, that the sequence looks artificial. The film lets the objection stand, and leaves the question of the signal's origin **open at the end**.

7. First-principles thinking framework

First-principles thinking means breaking a problem into its basic parts, sorting what is known from what is assumed, noting the limits that apply, and rebuilding an explanation using only what can be justified. It is a way of working under uncertainty. It does not guarantee an answer, and a rebuilt explanation is only as good as the basic parts it started from.

Child-friendly version (for the Thinking Tool card): *Take it apart. Sort what you saw from what you guessed. Rebuild it using only what you can back up.*

What it is not

Practice	How it differs
Asking "why?" repeatedly	Useful for finding assumptions, but it can loop forever. First principles stops at things that can be checked
Analogy	Borrows a known pattern ("it looks like a pulsar"). Fast, but it imports the other case's assumptions. This is the Thinking Tool for the Art and Expression Wormhole
Brainstorming	Generates options without ranking them by evidence
Hypothesis testing	Needs a rebuilt claim to test. First principles builds the claim and the test, then the test does the judging
Systems thinking	Maps how parts interact. First principles first decides what the parts are. Systems mapping is the Thinking Tool for Evolved Naturalism

The eight-step sequence

Each step is shown with the fictional signal on Vetra, a question a learner can ask, a way to make it visible, and the mistake to watch for.

1. State the observation

- **For a child:** What did the instrument actually record?
- **Signal example:** Three bursts, each a train of pulses, recorded on one receiver, with these times and strengths.
- **Question:** What is the plainest description that someone who was not there could check?
- **Activity / visual:** A bare strip of pulses on a dark screen, with no labels, no colour, no commentary.
- **Watch for:** Smuggling a conclusion into the description ("the message began at…").

2. Separate observation from interpretation

- **For a child:** What did we see, and what did we add?
- **Signal example:** "Pulses" are seen. "Primes", "melody" and "greeting" are added.
- **Question:** Which words in my description would still be true if nobody had ever thought of a message?
- **Activity / visual:** Two columns on a screen: SEEN and ADDED. Learners drag phrases between them.
- **Watch for:** Treating a vivid interpretation as if it were data because it feels certain.

3. Identify assumptions

- **For a child:** What must be true for my idea to work?
- **Signal example:** To call it a message we assume a sender, a purpose, and that the pattern is not chance.
- **Question:** What am I taking for granted that I have never checked?
- **Activity / visual:** Peel the claim like layers, each layer labelled "assumed" or "checked".
- **Watch for:** Assuming only the obvious ones. The hidden assumptions are usually about the instrument, the sample and ourselves.

4. Break the problem down

- **For a child:** What are the parts of the problem?
- **Signal example:** The receiver and its electronics, the radio noise around it, the star, the space between, and how the pulses were counted.
- **Question:** Which part could produce this without anyone sending anything?
- **Activity / visual:** An exploded diagram of the chain from star to screen.
- **Watch for:** Breaking it into parts but only examining the exciting one.

5. Establish constraints

- **For a child:** What is allowed, and what is missing?
- **Signal example:** Physics limits how fast a star can spin or flicker, and one instrument cannot rule out a fault in itself.
- **Question:** What explanations does known physics forbid, and what evidence do we not yet have?
- **Activity / visual:** A narrowing funnel: many explanations enter, constraints remove some.

- **Watch for:** Using "we do not know" to mean "anything is possible".

6. Generate competing hypotheses

- **For a child:** What else could explain this?
- **Signal example:** A spinning star; interference from the planet or a passing craft; a fault in the receiver; a natural flare; a deliberate signal.
- **Question:** Can I write at least three explanations that fit the same pulses?
- **Activity / visual:** Three suspect cards on a table, each with the clue that fits and the clue that does not.
- **Watch for:** Falling in love with the first explanation and treating the others as decoration.

7. Design a discriminating test

- **For a child:** What observation would make two explanations give different answers?
- **Signal example:** Point a second, independent instrument at Vetra. A natural source should show up on both; a fault should not.
- **Question:** If I am right, what should I see that I would not see if I am wrong?
- **Activity / visual:** A fork with two branches, each labelled with its prediction.
- **Watch for:** A test that every explanation would pass.

8. Update the explanation

- **For a child:** What changes when the evidence disagrees with my favourite idea?
- **Signal example:** If the second instrument sees nothing, the leading idea weakens. If both see it, the fault idea weakens.
- **Question:** What result would make me drop my idea?
- **Activity / visual:** A dial labelled "how sure am I?" that moves when new evidence is added.
- **Watch for:** Changing the test after the result so the idea survives.

Why this tool fits this story

The signal story is built so that steps 2 and 3 are where the adults fail and step 7 is where Ira's question leads. The sequence demonstrates that first-principles thinking is most useful when no answer is available, because it tells you what to do next.

8. Seven-minute episode: storyboard and narration

Working title: The Signal That Shouldn't Exist. **Target runtime:** 7:00 (brief allows 6:30–7:30). **Format:** animated worlds (Acts 1–4, 6) cut with live footage and one on-camera scientist (Act 5). Narration and dialogue total about 700 words at an unhurried pace, plus the scientist's 30–40 seconds, leaving more than two minutes for silence, music and image.

Runtime map

Act	Timestamp	Length	Job
1. Wonder	0:00–0:45	0:45	Make the planet irresistible
2. Possibility	0:45–1:40	0:55	Show how they live, and what the Kardashev scale does and does not measure
3. Disruption	1:40–2:40	1:00	Signal arrives, experts split
4. The overlooked assumption	2:40–4:15	1:35	Ira asks the question
5. The real frontier	4:15–6:05	1:50	Three real cases, one real scientist
6. The mission	6:05–7:00	0:55	Hand over the investigation

Sound motif. A single tuned pulse is introduced as part of the score in Act 1. In Act 4 the score drops out and only raw clicks remain. The viewer realises the "music" in the signal was supplied by the film, which is the lesson in miniature: framing shapes what you hear.

Labels on screen. From Act 3 onward, a small corner tag marks every claim as **ESTABLISHED**, **PLAUSIBLE SPECULATION** or **INVENTED**. This is the film's quiet way of teaching learners to ask which one they are looking at.

Act 1 — Wonder (0:00–0:45)

- **Narrative purpose:** make the viewer want to live on this planet.
- **Visuals:** black. A point of orange light grows into a sun. Pull back to a planet with two moons and a crowded night sky. Close on a city built like an open laboratory: children drawing equations in light on a wall, a musician tuning a telescope mirror, a girl carrying a ledger across a bridge.

- **Sound and music:** low sustained strings, no percussion. One tuned pulse enters at 0:30, soft, like a heartbeat.
- **Question introduced:** What would a civilisation of people who know everything be like?
- **Assumption challenged:** that being brilliant in many fields makes you right (planted, not yet challenged).
- **Thinking move modelled:** none yet. This is the setup.
- **Transition:** hard cut to the Commons.

> Imagine a planet where nobody has just one subject.
>
> A musician designs the spacecraft. An artist builds the model of the storm. And a twelve-year-old can walk up to the most celebrated scientist alive and tell her she's wrong, and the scientist will ask for her reasons.
>
> This is a planet of polymaths. They have solved more problems than we have names for.
>
> Hold on to that. It will matter.

Act 2 — Possibility (0:45–1:40)

- **Narrative purpose:** make the world coherent and prime the paradox.
- **Visuals:** the **Commons**, a public square where claims hang on glowing boards and anyone can strike a line through one. A mural that plots a star's light. A song that carries a dataset. Orbital collectors catching sunlight. Then a simple three-step graphic: planet, star, galaxy. A marker sits just below the first step on Anvaya's bar, and a lower one for Earth.
- **Sound and music:** strings widen. The pulse grows into a steady, gentle rhythm.
- **Question introduced:** What does a civilisation know about itself?
- **Assumption challenged:** that more power means more wisdom.
- **Thinking move modelled:** noticing what a measurement measures (power, not truth).
- **Transition:** the rhythm stops. Night. A receiver begins to count.
- **Tag on screen:** the scale is **ESTABLISHED** (as an idea). Anvaya is **INVENTED**.

> They gather most of the energy that reaches their world, and they've begun catching sunlight in orbit. Scientists here sort civilisations by that kind of power: a planet, a star, a galaxy. Anvaya is almost at the first step. We are not.
>
> But look at what that scale measures. Power. Not wisdom. Not whether you are right.
>
> Their schools are laboratories. Their art measures things. Their councils of mathematicians, musicians, biologists and engineers meet in public, where anyone can challenge any claim.
>
> By every measure we have, they are extraordinary.
>
> Then, one night, on the far side of their larger moon, a receiver starts to count.

Act 3 — Disruption (1:40–2:40)

- **Narrative purpose:** deliver the puzzle and make disagreement vivid.
- **Visuals: the Quiet Side**, a dish array on the lunar far side, shielded from the planet's radio noise. A screen: three bursts of pulses from a star called Vetra. The Council convenes. Four experts, four overlays: numbers glow for the mathematician, a staff of notes for the musician, a spinning-star model for the physicist, glyphs for the linguist. Overlapping voices build into noise.
- **Sound and music:** the pulse, now eerie. As each expert speaks, a different instrument lays its interpretation over the clicks.
- **Question introduced:** What is it?
- **Assumption challenged:** that a regular pattern must be meaningful.
- **Thinking move modelled:** Step 1 (state the observation). The facts are stated plainly: three bursts, one receiver, a variable star.
- **Transition:** voices stop mid-sentence. A child at the back of the room.
- **Tag on screen:** the signal is **INVENTED**.

> The signal comes from a star called Vetra. Three bursts, each a train of pulses, caught by one receiver and no other.
>
> The Council meets.

Mathematician: The spacing follows a sequence. Forty-seven pulses, then it breaks. That is no accident.

Musician: Listen. It has phrasing. It breathes.

Physicist: A spinning star would do this. Or something we haven't named.

Linguist: If we treat the first burst as a greeting…

> Four brilliant people. Four different patterns. And every one of them has already decided the same thing: that somebody is speaking.
>
> Cycles pass. They cannot agree.

Act 4 — The overlooked assumption (2:40–4:15)

- **Narrative purpose:** the turning point. Intelligence meets an untested assumption.
- **Visuals:** a quiet corner. **Ira**, twelve, reads the receiver's calibration logs on the night shift, tea going cold. She stands. The score drops out completely; only raw clicks remain. She walks to a wall board and splits it into two columns, **SEEN** and **ADDED**. "Pulses" goes under SEEN. "Message", "greeting", "melody" go under ADDED. The Council's words peel away from the signal one by one.
- **Sound and music:** silence under the clicks, then room tone. No score for 40 seconds.
- **Question introduced:** What if it is not a message at all?
- **Assumption challenged:** that the signal is meaningful before that has been shown.
- **Thinking moves modelled:** Steps 2 and 3 (separate observation from interpretation, identify assumptions), then the hint of Step 7 (what test could say no?).
- **Transition:** nobody speaks. The film lets the pause run, then cuts to Earth.
- **Tag on screen:** Ira's method is **ESTABLISHED** practice. The signal remains **INVENTED**.

> In the corner of the room, someone has been reading the instrument's logs instead.
>
> Ira is twelve. She keeps the receiver's calibration records on the night shift. She has no theory about what the signal says.
>
> She has a question no decoder was built to ask.

Ira: What if we decided it was a message before we checked that it was one?

> Every tool in that room was designed to translate. None of them could say: there is nothing here.
>
> Ira doesn't claim it is noise. She does something smaller and harder. She writes down what the receiver actually recorded, and beside it, everything the Council has added.
>
> Pulses: seen. Message: added.
>
> The room goes quiet. Not because she is right. Because nobody can say how they would know.
>
> The signal exists. It was measured. What shouldn't exist is certainty nobody has earned.

Physicist (after a beat): Then what would tell us?

Act 5 — The real scientific frontier (4:15–6:05)

- **Narrative purpose:** show that this is not just a story. Real scientists have stood in Ira's position.
- **Visuals:** cut to a real radio telescope at night. Archive footage and animated reconstructions of each case. Then a scientist on camera in their working space.
- **Sound and music:** the score returns, warmer and more grounded, with the pulse motif now played by a real instrument.
- **Question introduced:** Is this a real problem? How do real scientists decide?
- **Assumption challenged:** that the "alien or not" question has a quick answer.
- **Thinking moves modelled:** Step 6 (competing hypotheses), Step 7 (discriminating test), Step 8 (update). Each case shows one.
- **Transition:** the scientist poses the question to the viewer. Cut back to Anvaya.
- **Tags on screen:** each case is **ESTABLISHED**. That no technosignature has been confirmed is **ESTABLISHED** at the time of recording and must be rechecked before release (see section 12). The "search will continue" line is **PLAUSIBLE SPECULATION**.

> This has happened on our world.
>
> In 1967, a graduate student named Jocelyn Bell found radio pulses so regular that her team half-jokingly labelled the source LGM-1: Little Green Men. They did not announce a message. They looked for a second case, and found one, in a different part of the sky with a different rhythm. Today we call them pulsars: spinning neutron stars.
>
> In 2015, odd bursts at Australia's Parkes telescope were traced to a microwave oven in the staff kitchen, opened a few seconds early.
>
> In 2021, a candidate signal seen while pointing at Proxima Centauri, the nearest star beyond our Sun, was judged to be human interference.
>
> Then there is the Wow! signal, from 1977. Seventy-two seconds. Never seen again. No explanation has been confirmed.

Scientist on camera (30–40 seconds, in their own words; prompts below):

- What do you do when a signal looks artificial?
- What would make you say this is a message?
- What have you been wrong about?

> Looking for signs of technology elsewhere is called searching for technosignatures. As far as anyone has shown, none has been confirmed. The search goes on. What changes the answer is not a stronger hope. It is a better test.

Act 6 — The mission (6:05–7:00)

- **Narrative purpose:** hand over the investigation.
- **Visuals:** back on the Quiet Side. Ira slides the logbook toward the camera. A single empty seat is lit at the Council table. A fact card appears: *three bursts · forty-seven pulses, then a break · one receiver · a variable star.* The four prompts appear one at a time. Final card: *Message, or not a message? You have two minutes.* Then the brand lines.
- **Sound and music:** the pulse, unresolved. It does not end on a chord.
- **Question introduced:** What do you think is happening, and how would you find out?
- **Assumption challenged:** none new. The viewer is asked to apply the one they have just seen.

- **Thinking move modelled:** all eight, handed over.
- **Transition:** to the Spark, then the Reply Card (section 9).

> Anvaya is a world that does not exist. The question is real: when something strange arrives, how do you decide what it means?
>
> The Council has one empty seat.
>
> Here is all they know. Three bursts. Forty-seven pulses, then a break. One receiver. A star that changes its brightness.
>
> Tell us what you think is happening. What else could explain the same clues? What would you measure to tell them apart? And what would change your mind?
>
> Draw it. Write it. Build it. Record it. No telescope needed.
>
> First, pick a side. Message, or not a message. You have two minutes.
>
> Nobody knows yet.

End card: Asymmetry, A Cloud School. by Naxxatra. *Learn what nobody knows yet.*

Accessibility and versions

- English and Hindi at launch, as the brief requires. Recommended: write the Hindi narration separately, not by dubbing, because lines such as "Pulses: seen. Message: added." rely on rhythm.
- Open captions in both languages. A low-bandwidth cut (audio plus stills) of the same film.
- No Act depends on colour alone. The SEEN and ADDED columns are labelled by text.

9. The learner's first mission

Name: The Signal That Shouldn't Exist — Investigator's Mission. In Asymmetry's vocabulary this is the **Reply Card**: a child's reply to the episode, which is also their application. The brief's journey has two steps here, and both are designed below.

Step 1: the Spark (two minutes)

Immediately after the film, the viewer picks a side with one tap or one circled word, and writes one sentence why.

- **Message**
- **Not a message**
- **I can't tell yet** (a fully valid answer)

The Spark is not scored. It exists to commit a child to a position they can later be asked to defend, doubt or change. After the Reply, the learner sees how other viewers answered. The point is that the split is real.

Step 2: the Reply Card (learner-facing brief)

This is the text to publish beside the video and print on the school Reply Card. It is written for a 12-year-old and meant to be readable by a parent.

> **The Council has one empty seat. It is yours.**
>
> **What we know.** Three bursts of pulses from a star called Vetra. Forty-seven pulses, then the pattern breaks. One receiver saw it. The star changes its brightness.
>
> **Your job.** Work out what could be happening, and how we could find out. You can use any of these questions:
>
> 1. What do you think is happening?
> 2. What else could explain the same clues?
> 3. What would you need to know?
> 4. What would you look for to tell your ideas apart?
> 5. What would make you change your mind?
>
> **Show us any way you like.** A drawing with labels. A short story. A model made from things at home. A voice note. A one-minute video. A page of writing. An experiment you could run. A maths argument.
>
> **You do not need** a telescope, coding, advanced maths, or to know anything about astronomy.
>
> **How long?** As long as it takes you to think well. A single page, a minute of audio, or a minute of video is plenty.
>
> **Language.** English or Hindi, or a mix. We care about your thinking, not your grammar or accent.
>
> **What happens next.** Every Reply is read. After two weeks, the episode's scientist answers three Replies on video. If you would like to keep going, a live Discovery session with a mentor is the next step.

Why this mission works as an application

- **It tests reasoning, not recall.** Nothing in it can be looked up. A learner with no astronomy background and a learner with a telescope start level.
- **It has no right answer.** The signal's origin is deliberately open, so there is nothing to copy.
- **It reveals how a child handles being wrong.** The last question asks what would change their mind. A child who cannot answer it has told the reviewer something useful.
- **It works across formats and bandwidths.** A photographed drawing, a short voice note and a printed card all carry the same thinking.

Example Replies (illustrative; not real learner work)

These show that different strengths can be equally strong. They are for reviewer calibration (section 10).

Explorer or Fellow	Format	What stands out
Explorer, 11	Hand-drawn comic	Shows a character checking the receiver's own wires first, and a panel that says "If a second telescope hears nothing, I was wrong." Plain drawing, clear reasoning
Explorer, 13	Voice note, 90 seconds	Names three explanations, says the star's brightness changes might explain the bursts, and asks "What did the star do on the nights when there was no signal?"
Fellow, 15	One page of maths	Argues that 47 pulses before a break could occur by chance, estimates roughly how often, and says what extra data would change the estimate
Fellow, 16	Experiment proposal	Proposes pointing a second instrument at Vetra and at a nearby star as a control, and lists what each outcome would mean
Explorer, 10	A short story	The story ends before the answer. The child has written what the narrator would need to find out, and it is a better list than the Council's

10. Application and review framework

The brief fixes four public criteria: curiosity, reasoning, honesty and agency, "never on polish, accent or English fluency". This framework keeps those four and places the six areas from the design brief beneath them as reviewer prompts, so nothing the learner sees is more complicated than four words.

How the six areas map to the four criteria

Public criterion	Reviewer prompts drawn from the six areas
Curiosity	Question formation: did they ask something new, or something the film did not ask? Originality and imagination: is there a surprising angle, explanation or image?
Reasoning	Use of evidence: do they separate what was seen from what is guessed? Alternatives: do they offer at least two explanations that fit the same clues? Do they propose something that would tell them apart?
Honesty	Willingness to consider alternatives: do they say what would change their mind, and what they do not know? Do they avoid claiming more than the clues allow?
Agency	Initiative and follow-through: did they choose their own approach, and carry it through to something concrete? Communication is judged only as "can a reviewer follow the thinking", never as presentation quality

What each criterion looks like, by age division

A reviewer is looking for evidence of a habit, not a finished skill. The same four criteria apply to both divisions, with different expectations.

Criterion	Signals in an Explorer (10–13)	Signals in a Fellow (14–17)
Curiosity	Asks a question the film left out. Wonders about the star, the instrument or the sender	Asks a question that opens a line of investigation and is aware of what would be required to answer it
Reasoning	Names two ideas. Says why one clue fits one idea better	Treats ideas as testable. Gives each a prediction. Uses a number, a diagram or an argument

Criterion	Signals in an Explorer (10–13)	Signals in a Fellow (14–17)
Honesty	Says "I don't know" or "I might be wrong if…"	Names a specific result that would refute their favourite idea. Acknowledges what the data cannot show
Agency	Makes something themselves, in their own way	Selects a method, justifies the choice, and anticipates a limit of their own approach

Four levels, no pass or fail

Each criterion is read at one of four levels: **Not yet visible**, **Emerging**, **Clear**, **Distinctive**. A Reply produces a profile across the four criteria, not a total. This lets reviewers recognise a learner who is Distinctive in curiosity and Emerging in reasoning, and route them to the right Studio.

How reviewers avoid favouring polish or a single format

1. **Format-neutral inputs.** Reviewers read a transcript or description, plus the original. Audio and video are transcribed. Drawings are described by the learner in one sentence.
2. **Blind first read.** Name, school, city, language and age division are hidden on the first read. Reviewers see only the Reply.
3. **Two independent reviewers.** A third is brought in when profiles differ by two or more levels on any criterion.
4. **A calibration set.** Before each review cycle, reviewers score the same five to ten example Replies (including those in section 9) and discuss differences.
5. **Score the thinking, not the finish.** A messy drawing with a good reasoning chain beats a beautiful drawing of one idea.
6. **Same criteria in any language.** Hindi and English Replies are reviewed by reviewers fluent in the language used, against the same descriptors.
7. **Spot checks for polish bias.** Periodically compare profiles against format, language, school type and region. If any group is consistently lower, review the descriptors, not the children.

Checking that the thinking is the child's own

A learner may get help from a parent, teacher or an AI tool. The mission is built to make this less damaging: the questions ask for personal judgement ("which side would you pick, and what would change your mind?"), and the next step is a live Discovery session. A short, friendly conversation about the child's Reply is the verification. It is not a trap. It is a chance for the child to explain their idea in their own words.

What a review leads to

Outcome	What it means
Reply acknowledged	Every child who replies gets a response that names one thing they did well
Featured	A small number are shown (with permission) as examples of different strengths. The brief commits to a scientist answering three Replies on video after two weeks. Selection should favour variety of approach, not top scores
Invited to a Discovery session	The next step in the brief. Invitation is based on evidence of the four criteria, not on a total
Level and need-based places	Decided separately from the Reply review, so ability to pay never enters the scoring and scoring never replaces need assessment

Decision to confirm: whether there is a minimum profile for a Discovery session or whether every Reply is invited. Section 15 sets out the trade-offs.

11. The connected planetary universe

The seven worlds are imaginative environments, not real planets and not astronomical discoveries. They are designed as one system: each world needs something that another holds, and each carries a real question underneath. The Polymath Planet comes first because it teaches the habit the others depend on.

Improvements to the proposed set

- **Planet of Justice:** reframed so that it is a society that has built perfect oversight and must decide whether that makes it fair. Nothing in it involves exile. The real frontier is how rules for shared spaces (the Antarctic, the deep sea, orbit) get made.
- **Planet of Play:** the central question is kept, and the real frontier is sharpened: game theory, the evolution of cooperation, optimisation and what happens when a system is optimised for one measure.
- **A shared thread:** every planet receives the Vetra signal, or a consequence of it, in its own way. This makes the network a single story and not a tour.
- **A final "mirror" stage (recommended, optional):** after the seven worlds, learners turn the lens on Earth: what does the view from outside tell us about home? This is where the Earth Wormhole connects.

The seven worlds

Planet of Polymaths

- **Premise:** a civilisation where nobody has one subject (section 6).
- **Human concern:** expertise, and its blind spots.
- **Real frontier:** interdisciplinary science, and how researchers audit their own assumptions.
- **Central question:** Does knowing more necessarily mean understanding better?
- **Disciplines:** mathematics, physics, philosophy of science, statistics.
- **Learner mission:** the Reply Card (section 9).
- **Connects to:** every other planet, because each depends on its habits.
- **Advances the Wormhole:** teaches the central tool and sets the rule that every claim is challenged.

Planet of Artists

- **Premise:** a world where art is a measuring instrument. Paintings record a storm's pressure, music carries a genome.
- **Human concern:** how to share what we cannot yet explain.
- **Real frontier:** interstellar messages (Voyager Golden Record, radio messages sent from Earth), data sonification and visualisation, and the debate about whether to transmit at all.
- **Central question:** Can we communicate something we cannot yet explain?

- **Disciplines:** visual art, music, semiotics, data visualisation, linguistics.
- **Learner mission:** design a message for a reader who shares none of your language or senses. Then list three ways it could be misunderstood.
- **Connects to:** Polymaths (they cannot read the signal without translators), Stars (artists turn raw data into images).
- **Advances the Wormhole:** shows that representation is not neutral, which deepens the lesson of the film's silent score.

Planet of Invention

- **Premise:** a civilisation that can build almost anything and has failed to build what it most needs.
- **Human concern:** unintended consequences, and building for the wrong problem.
- **Real frontier:** life support and recycling in closed systems, making resources from local materials, space debris, and reliability engineering.
- **Central question:** Can a civilisation invent everything and still fail to solve its most important problems?
- **Disciplines:** engineering, chemistry, materials science, systems design.
- **Learner mission:** design a closed-loop air or water system for a habitat of ten people, and write its three most likely failure modes.
- **Connects to:** Life (what must the system keep alive?), Justice (who controls it?).
- **Advances the Wormhole:** supplies the "how" for the beyond-Earth project, with an honest list of what does not yet work.

Planet of Justice

- **Premise:** a society in which everything is seen and recorded, so rules are enforced perfectly. Is it fair?
- **Human concern:** freedom, fairness, accountability and who decides.
- **Real frontier:** international space law, shared-resource governance, and the ethics of enforcement in an enclosed habitat where nobody can leave.
- **Central question:** Is a perfectly controlled society necessarily a just society?
- **Disciplines:** ethics, law, political philosophy, social science, data ethics.
- **Learner mission:** write the first three rules of a settlement, and a way for its people to change them.

- **Connects to:** Invention (life-critical systems raise hard questions about control), Life (what do we owe other life?).
- **Advances the Wormhole:** supplies the governance layer of the culminating project.

Planet of Play

- **Premise:** a society where every game is won by the best-optimised player, so winning is certain.
- **Human concern:** meaning, competition and cooperation.
- **Real frontier:** game theory, the evolution of cooperation, optimisation and its side effects, and how sport and exercise change in low gravity.
- **Central question:** If winning becomes guaranteed, does a game remain meaningful?
- **Disciplines:** mathematics, behavioural science, sports science, computer science.
- **Learner mission:** design a game where cooperation is the only way to win, and test it with three people.
- **Connects to:** Justice (rules and fairness), Invention (what happens when a metric becomes the goal).
- **Advances the Wormhole:** gives learners a way to model decisions under uncertainty, and hands the settlement project its first stress tests.

Planet of Life

- **Premise:** a world where living things share nothing with Earth's biochemistry, and scientists argue about whether some of them are alive.
- **Human concern:** recognising something unlike ourselves.
- **Real frontier:** astrobiology, extreme environments, definitions of life, and how to evaluate a claimed biosignature.
- **Central question:** How would we recognise life if it looked nothing like life on Earth?
- **Disciplines:** biology, chemistry, geology, planetary science, philosophy.
- **Learner mission:** design a decision tree for an unknown sample: what would you test, and what non-living process could fake each result?
- **Connects to:** Stars (technosignatures vs biosignatures), Justice (what we owe other life).
- **Advances the Wormhole:** gives Question A its biological half.

Planet of Stars

- **Premise:** a world of observers whose only knowledge of the universe is light. They must decide what an advanced civilisation would leave behind.
- **Human concern:** inference from distance and incomplete evidence.
- **Real frontier:** technosignatures, the Kardashev scale, and known cases where an odd stellar signal turned out to have a natural explanation.
- **Central question:** What could reveal the existence of an advanced civilisation?
- **Disciplines:** astronomy, physics, data science, optics.
- **Learner mission:** pick one possible sign of technology, then say how nature could produce the same sign and what extra observation would separate them.
- **Connects to:** Polymaths (the Vetra signal comes from here), Artists (data into pictures), Life (comparing the two kinds of search).
- **Advances the Wormhole:** gives Question A its technological half.

How the worlds depend on one another

From	To	What they supply
Polymaths	All	The habit of testing claims, and the Reply method
Stars	Polymaths, Artists	Raw data about Vetra and how to read it
Artists	Stars, Polymaths	Ways of showing data that do not mislead
Life	Invention, Justice	What a habitat must keep alive and what we must not harm
Invention	Life, Justice	Systems that keep people alive, and their failure modes
Justice	Invention, Play	Rules that decide who controls those systems
Play	Justice, Invention	Models of cooperation and incentives under stress

The network makes a single point: no major question here is answerable from one planet. A settlement needs engineers and ethicists, a signal needs artists and statisticians, and the people who study life need the people who make rules.

12. Real scientific frontiers and research sources

Every fictional premise in this Wormhole points at a real open question. The fiction is not evidence: it is a door. This table shows what lies behind each door, what is settled, what is not, and where a learner can start. Searches were run on 16 October 2026 against NASA, ESA, COSPAR, UNOOSA, the Berkeley SETI Research Center and peer-reviewed literature. The links below came from those searches. They have not all been read in full, so every claim needs subject-matter review before publication (section 14).

Status labels used: Established (well supported, as of the source date), **Contested** (published, disputed), **Open** (no agreed answer), **Speculative** (an idea without evidence).

What is not claimed anywhere in this Wormhole: that extraterrestrial life or technology has been found. As far as these sources show, no technosignature has been confirmed, and no biosignature has been confirmed on any world.

#	Area and fictional link	Established	Open, contested or speculative	Question for ages 10–17	Investigation or prototype	Sources
1	**Origin and conditions of life** (Planet of Life). Chemistry, biology, geology	NASA offers the Ladder of Life Detection as a tool for judging claims of life, built from lessons of earlier attempts	**Open:** how non-living chemistry became life, and whether that is common	What is the shortest list of things something needs in order to count as alive?	Test fire, crystals, a mould and a snowflake against your own list. Where does your list break?	NASA Ladder of Life Detection; Neveu et al. 2018, Astrobiology
2	**Habitability beyond Earth** (Planet of Life, Planet of Stars). Astronomy, planetary science	The NASA Exoplanet Archive listed 6,147 confirmed planets on 12 March 2026. Habitable-zone means conditions that could allow liquid water, not that anything lives there	**Contested:** a 2023 and 2025 claim of dimethyl sulfide on K2-18 b was followed by reanalyses finding no conclusive evidence. **Open:** whether any known planet is inhabited	If a planet sits in the habitable zone, what else must be true before you'd call it a home for life?	Sort a public exoplanet table by star type and orbit, shortlist three, and list two extra checks for each	NASA Exoplanet Archive news; Astronomy.com on K2-18 b

#	Area and fictional link	Established	Open, contested or speculative	Question for ages 10–17	Investigation or prototype	Sources
3	**Biosignatures and technosignatures** (Planet of Life, Planet of Stars). Chemistry, statistics, instrumentation	Scientists use confidence scales such as the Confidence of Life Detection (CoLD) scale: first a detection, then ruling out non-living sources, then independent confirmation. In September 2025 NASA reported a "potential biosignature" in a Mars rock sample, with confirmation needing samples returned to Earth	**Hypothesis:** the Mars result. **Open:** whether sample return will happen, and what it will show	What would you have to rule out before you said "this is life"?	A "false-positive detective" game: a list of mystery readings, and for each, three non-living explanations	Scientific American on the CoLD scale; Sky at Night on the Perseverance finding
4	**Search for extraterrestrial intelligence** (Polymath Planet, the Signal). Radio astronomy, signal processing	Pulsars (1967): regular pulses first nicknamed LGM-1, later explained by spinning neutron stars. Perytons (2015): bursts at the Parkes telescope traced to a microwave oven door opened early. BLC1 (2021): a Proxima Centauri candidate judged human interference	**Open:** the Wow! signal (1977, 72 seconds, never recurred). A 2024 analysis proposed a hydrogen-cloud explanation, but none is confirmed. As far as these sources show, no technosignature is confirmed	Why do scientists insist on a second telescope?	Record the same noisy sound on two devices and see which features are shared. Add an AM radio near a switching appliance (adult supervised) to see interference	Bell Burnell and pulsars, APS; Petroff et al. 2015; Breakthrough Listen BLC1; Wow! signal analysis, arXiv
5	**Planetary environments and extreme habitats** (Planet of Life). Geology, microbiology	Life on Earth survives in places once thought impossible, such as hot vents, ice and deep rock	**Open:** the true limits of life, and whether other worlds' oceans or rock host any	Where on Earth looks most like another world, and what lives there?	An "extreme habitat postcard" for one Earth site and one other world, with what would need to be true for life in each	NASA Astrobiology

#	Area and fictional link	Established	Open, contested or speculative	Question for ages 10–17	Investigation or prototype	Sources
6	**Energy and technological civilisations** (Planet of Stars, Polymath Planet). Physics, astronomy	The Kardashev scale ranks civilisations by energy use as an idea. Boyajian's Star showed irregular dimming that once prompted megastructure speculation; later observations of how the dimming varied by wavelength favoured dust	**Speculative:** Dyson-type megastructures. **Open:** the exact cause of the star's dimming	How much energy does your city use compared with the sunlight falling on it?	A Fermi estimate: Earth's energy use versus the sunlight it receives, and what step on the scale that implies	GeekWire on the dust finding; Scientific American on Boyajian's Star
7	**Human health and long-duration spaceflight** (Planet of Invention, beyond-Earth project). Physiology, physics, medicine	NASA's Twins Study followed one astronaut over 340 days on the ISS and found health "mostly sustained", with caveats: one pair, and still inside Earth's magnetic shield. Curiosity's radiation detector measured about 1.8 mSv a day on the way to Mars, roughly 0.66 Sv for a round-trip transit with current propulsion	**Open:** the long-term effect of deep-space radiation and partial gravity. Career dose limits are set and revised by agencies, so check the current figure	What does your body do when it stops carrying its own weight?	Design a daily exercise and shielding plan for a six-person crew, listing what each choice costs in mass and time	NASA Twins Study results, CNN summary; Mars transit radiation, National Geographic

#	Area and fictional link	Established	Open, contested or speculative	Question for ages 10–17	Investigation or prototype	Sources
8	**Closed-loop ecosystems and off-world settlements** (Planet of Invention, Planet of Life). Ecology, chemistry, engineering	The ISS needs regular resupply. Biosphere 2 was the largest closed ecological facility built and showed how hard such systems are to control. ESA's MELiSSA programme works towards a closed life-support loop with close to 100% efficiency	**Open:** whether a small closed ecosystem can stay stable for decades. Per the sources found, no artificial closed system has supported humans over long periods without outside input	What happens to a sealed ecosystem when one part fails?	A sealed jar garden for four weeks, with a daily log and a failure post-mortem	ESA MELiSSA; Biosphere 2's lessons, Space: Science and Technology
9	**Ethics of exploration and planetary protection** (Planet of Justice, Planet of Life). Law, ethics, microbiology	The Outer Space Treaty (1967), Article IX, calls for avoiding harmful contamination of celestial bodies. COSPAR maintains the international reference policy on planetary protection, last updated in 2020. The Artemis Accords are non-binding principles on safety zones, heritage and space resources	**Contested:** how the Accords square with the Outer Space Treaty's ban on national appropriation. **Open:** how any of this applies to a permanent settlement	If microbes might live on Mars, should humans land there at all?	A mock negotiation: four teams (science, industry, settlers, a possible Martian biosphere's advocates) draft one clause	COSPAR Planetary Protection Policy; NASA on the Artemis Accords; ASIL analysis
10	**The future relationship between humanity and Earth** (the mirror stage). Earth science, ethics	Satellites already track forests, water and weather on a global scale	**Contested:** whether space settlement helps or distracts from protecting Earth	What can you see about Earth from space that you cannot from the ground?	Use free public satellite imagery to follow one place over time and report what changed. The Earth Wormhole picks this up	Link to confirm: NASA Earth observation tools

Two cautions for reviewers. First, counts and statuses change: the exoplanet total and the Mars sample-return plans should be rechecked on the day of release. Second, a source being on this list does not mean the learner-facing wording is cleared: each Act 5 statement and each table cell needs sign-off by a working scientist.

13. The culminating beyond-Earth project

Working concept: Design a civilisation that could survive beyond Earth, and decide what would make it worth building. It is a Studio-length team project (one term at minimum) that can be extended across three terms for a Frontier Fellow research track presented at Frontiers.

The project is deliberately two-sided. A team must show that a settlement could work, and then decide whether it should exist. A team that concludes it should not is doing the project as intended.

Questions the project must answer

- What conditions are needed for a settlement to survive, and which are hardest?
- What must be made locally, and what must come from Earth?
- How do energy, food, water, air and waste fit together?
- Which physical limits (radiation, gravity, distance, supply) set the design?
- How is the settlement governed, and how can its rules change?
- How do its people decide when they are uncertain?
- What do they owe any life they might find, and Earth, and each other?
- What from Earth would they choose to keep?
- Under what conditions is leaving Earth justified, desirable or unnecessary?

The dossier: what every team produces

A team may choose its form (see next table), but the dossier must contain each of these:

1. **Site and constraints sheet:** where, and what that place does to humans (radiation, gravity, temperature, resources). Cites real data.
2. **Survival model:** rough numbers for energy, water, food, air and waste for a stated crew size, with a Fermi-style estimate and a stated margin of error.
3. **Local versus imported:** a list, with what happens if imports stop.

4. **Governance:** a short constitution, how it is amended, how disputes are settled, and who can say no.
5. **Failure plan:** three ways the settlement could fail and what its people do.
6. **Obligations:** a planetary-protection plan for the site (what the team would and would not do if signs of life were found), and what the settlement owes Earth.
7. **What we keep:** a short list of what the settlement carries from Earth, and why.
8. **The position:** the team argues one of three, and gives the strongest case against itself:
   - **Go:** leaving Earth is justified.
   - **Do not go:** the effort and risk are better spent elsewhere.
   - **Go only if:** it is justified under named conditions.
9. **Uncertainty register:** what the team does not know, and which unknowns would change its mind.

Many outputs, one project

Output	Best suited to a learner who…
Habitat prototype or scale model	builds, tests and breaks things
Digital simulation (a spreadsheet counts)	likes numbers and systems
Planetary settlement model	enjoys mapping and logistics
Research dossier	enjoys reading, comparing sources and writing
Fictional constitution	cares about fairness, language and rules
Resource-allocation system	likes designing incentives
Illustrated field guide to the site	thinks through drawing and observation
Collaborative mission plan with a timeline	likes coordinating people

A team's dossier needs only one polished output. The others can be sketches.

Roles: different strengths, one outcome

Strength	What they contribute	Where it connects
Maker	Prototype of a failure-critical system (air, water, shielding)	Planet of Invention
Modeller	Resource and risk numbers, sensitivity checks	Planet of Stars, Planet of Play
Biologist	What lives in the habitat, and what must not be disturbed outside	Planet of Life
Ethicist or lawyer	Governance, rights, planetary-protection plan	Planet of Justice
Storyteller or artist	How the settlement explains itself to Earth and to its own children	Planet of Artists
Red-teamer	Attacks another team's dossier and its weakest assumption	Polymath Planet

The red-team role matters most. Each team's dossier is handed to another team, who must try to find the hidden assumption. This is where first-principles thinking returns at scale.

How it runs across three terms (Fellow track)

Term	Focus	Main deliverable
1	**Survive.** Site, constraints, life-support loops, failure modes	Constraints sheet and survival model
2	**Decide.** Governance, incentives, decisions under uncertainty, obligations	Constitution, failure plan, planetary-protection plan
3	**Justify.** The position, red-team round, public communication	Full dossier and a Frontiers presentation

An Explorer version runs in one term with a given site, a fixed crew of 20, and a lighter survival model.

How it is judged

A panel of four (a scientist, an engineer, an ethicist and an artist or designer) uses the same four criteria as the Reply (curiosity, reasoning, honesty, agency), plus one project criterion: **is the design honest about what it cannot yet do?** A team that names its weakest assumption scores higher than a team that hides it.

Returning to the two questions

What lives in the deep universe? The project does not answer this. It asks teams what they would do if they found something, and what they would need to know before saying they had. The planetary-protection plan forces them to take the possibility seriously without assuming it.

What does leaving Earth mean for humanity? The project does not decide this either. It asks teams to show what survival would require, what it would cost, and who would decide. The value lies in the reasoning each team leaves behind, including the teams that end up arguing for staying.

14. Production and implementation roadmap

Build the episode and the Reply first, test them with a small group, and only then build the pathways. No platform work should begin until the first test shows that children produce original questions. Durations are indicative; the team should set dates against its own calendar and the brief's term structure.

Staged plan

Stage	What happens	Output	Gate to the next stage (evidence required)
1. Validate	Confirm the premise and objective with 3–5 educators and 2 working scientists. Pick the Act 5 scientist	One-page brief, signed-off premise	Scientists agree the framing of signals and the case studies are accurate

Stage	What happens	Output	Gate to the next stage (evidence required)
2. Script and storyboard	Turn section 8 into a locked script and a drawn storyboard. Write the Hindi version separately	Script, storyboard, animatic with temp voice	Read aloud to 5 children aged 10–17: they can restate the central question without prompting
3. Develop the Reply	Finalise the Reply Card, the Spark, the reviewer guide and example Replies	Reply Card in print, web and audio-only forms	A reviewer can score 5 sample Replies, and two reviewers agree within one level on 4 of 5
4. Test	Run the animatic and Reply with 20–30 learners from varied schools, regions and languages	Raw Replies, observation notes, interviews	At least some Replies contain an alternative explanation, and some name what would change their mind
5. Examine responses	Read all Replies for variety, quality and bias (section 10, spot checks)	Short findings memo	No format, language or school type is systematically scored lower for the same thinking
6. Refine	Revise story, narration, prompts and examples. Shoot the scientist segment	Locked cut	Test-group Replies improve on the second round, and the Spark shows a real split between sides
7. Build pathways	Draft the first Studio outlines from section 11, and the Dark Sky and Frontier Camp links from the brief	Three Studio outlines, one teacher kit	A Studio mentor can run the first session from the outline with no extra briefing
8. Develop the culminating project	Prototype the dossier and red-team round with one cohort of Fellows	Dossier template, panel guide	Two teams argue opposite positions with strong evidence

Essential versus optional

Essential	Optional (after validation)
Seven-minute film in English and Hindi, with captions	A full set of planet films
Printable Reply Card and web form, with audio and photo upload	Interactive web explorer of the signal
Reviewer guide and calibration set	Live reviewer dashboard
Scientist segment	Scientist video answering three Replies (brief requires it, so plan for it after the two-week window)
Thinking Tool card for first principles	Animated versions of each of the eight steps
Teacher lesson kit (the brief promises Wormholes as ready lessons)	Class-level thinking report

What can be made simply, and what needs expert review

Can be made with simple illustration, animation and web pages	Needs subject-matter review or more research
Anvaya, the Commons, the Quiet Side, Ira and the Council (stylised 2D)	Every statement in Act 5, and every row of section 12
The SEEN and ADDED board, the signal trace and the Kardashev graphic	The signal's design: it must be plausible without implying a real catalogued source
The Reply Card, Spark and thinking cards	The "no confirmed technosignature" statement, rechecked on release day
Audio-only and low-bandwidth cuts	The Hindi translation of scientific terms
	Consent and data handling for children's submissions
	Any claim about NASA, ESA or named researchers, and permission to use footage

Minimum testable version

A rough animatic with temp voice, one recorded scientist answer, a paper Reply Card, and a reviewer guide. Show it to 20–30 children in a classroom or a single online session. No branding polish, no platform.

Feedback questions for the test

For learners, straight after the film:

- Which moment surprised you most?
- What do you think was happening with the signal? What else could it be?
- What would you want to find out next?

After the Reply:

- How did you decide what to put in?
- Was there a question you wanted to ask but could not fit?
- Would you want to keep working on this? What would you want to do?

For teachers and parents:

- What did the child say or do after the film that surprised you?
- Did the Reply Card feel like a test? Why, or why not?

Evidence that the Wormhole is working

Views and likes are not evidence. Look for:

- **Original questions** that were not in the film.
- **Alternative explanations**, at least two per Reply from a meaningful share of learners.
- **Named tests:** Replies that say what they would measure to tell ideas apart.
- **Changed minds:** Replies that say what would change their mind, and learners who revise after the scientist's answer.
- **Independent follow-through:** children who investigate something on their own before any Studio.
- **A wish to continue:** requests for the next step, including from children who did not do well on any polish measure.
- **Spread:** responses from varied regions, schools, languages and ages, not only the best-resourced.

15. Open design decisions, risks and recommendations

Six decisions need an owner before scripting locks. Each is set out with options, trade-offs and a recommended direction.

Decisions

#	Decision	Options and trade-offs	Recommendation
1	**Who fronts Act 5**	(a) A radio astronomer or SETI researcher working in India: strongest link to the audience, narrower pool. (b) An international figure tied to one of the real cases: more name recognition, less local. (c) An early-career researcher: closer to the viewers' age and to the BLC1 story, where an undergraduate first spotted the signal. Institutions worth approaching (not yet contacted): NCRA-TIFR and the Raman Research Institute	Prefer (c) or (a). Choose someone who can say "I was wrong about…" on camera, because honesty is the Wormhole's tone
2	**Episode title**	Keep "The Signal That Shouldn't Exist" (strong hook, risk of implying the signal is impossible) or change to a neutral title	Keep it, and make Act 4's last line pay it off. Test the title with children before release
3	**Hindi version**	Dub the English, or write separately	Write separately, and have the Hindi reviewer check each scientific term

#	Decision	Options and trade-offs	Recommendation
4	**Minimum profile for a Discovery session**	A threshold keeps sessions focused but risks excluding late bloomers. Inviting everyone is fairer but may exceed capacity	Invite every Reply that shows at least Emerging in two of the four criteria. Review the threshold after the first cycle. Capacity is not stated in the brief, so confirm it
5	**Showing other viewers' Spark answers**	Show before the Reply (social proof, anchoring) or after (cleaner thinking)	Show only after the Reply is submitted
6	**Whether to add the "mirror" stage** (Earth seen from outside)	Adds a link to the Earth Wormhole but lengthens the path	Add it as an optional last stage, and decide after the first Studio test

Risks

Risk	Why it matters	Mitigation
Children come away thinking aliens are likely, or that SETI is pointless	The film's real cases mostly end as interference. That could read as "it's always nothing"	Keep the Wow! signal unresolved in the narration. Say in the scientist segment that the search is rigorous and ongoing
Fiction mistaken for fact	A fictional planet and signal sit beside real cases	Use the ESTABLISHED, PLAUSIBLE SPECULATION and INVENTED tags consistently. Subject-matter review of Act 5
Science errors in a public film	Counts, statuses and dates change	Recheck on release day. Maintain a sourced fact sheet. Keep claims in the film conservative
Polish bias in the Reply	Children with better devices or tutors may look stronger	Blind first read, transcribed audio, calibration set, spot checks (section 10)

Risk	Why it matters	Mitigation
Outside help on Replies, including AI tools	A polished Reply may hide thin thinking	Ask for "what would change your mind", and use a live Discovery session to talk through the Reply. Treat it as a conversation, not an interrogation
Children's data and consent	The Reply collects work, voice and possibly video from under-18s	Legal review of obligations under India's Digital Personal Data Protection Act, 2023, including parental consent, and equivalent rules in other regions served. Collect the minimum personal data
Reviewer capacity	Every Reply read, plus a scientist answering three, needs steady staffing	Pilot at small volume first. Size the review team from test data
Scientist selection of three Replies	Risk of favouring the polished	Select for variety of approach, as section 10 sets out
Token diversity	A Sanskrit-derived planet name and a girl lead can feel decorative	Test with learners from different regions and communities. Allow the planet's name and characters to change
Overloading the film	Seven minutes carries six acts	Cut Act 2 detail first. Protect Acts 4 and 6

Recommended next steps

1. Confirm the premise with two working scientists, and choose the Act 5 scientist.
2. Lock the script, with the Hindi version written separately.
3. Build the animatic and the paper Reply Card in parallel.
4. Run a test with 20–30 learners, and read every Reply.
5. Decide the Discovery-session threshold and the consent approach before the public release.

The design holds together as long as the film stays honest about what nobody knows yet. Everything else here can change.